

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

TRAINING TR-102 REPORT DAY 18

16 July 2025

Overview:

In today's session, we learned about Sentiment Analysis, a popular task in Natural Language Processing. It is used to detect the emotional tone behind a body of text such as positive, negative, or neutral opinions. This technique is widely used in analyzing reviews, social media content, and customer feedback.

Learning Objectives:

- Understand what Sentiment Analysis is and why it's important
- Learn how to preprocess and clean text data
- Explore how models are trained to classify sentiment
- See real-life applications and datasets like movie reviews or tweets

What is Sentiment Analysis?

Sentiment analysis is the process of classifying whether a block of text is positive, negative, or neutral. The goal that Sentiment mining tries to gain is to be analyzed people's opinions in a way that can help businesses expand. It focuses not only on polarity (positive, negative & neutral) but also on emotions (happy, sad, angry, etc.). It uses various Natural Language Processing algorithms such as Rule-based, Automatic, and Hybrid.

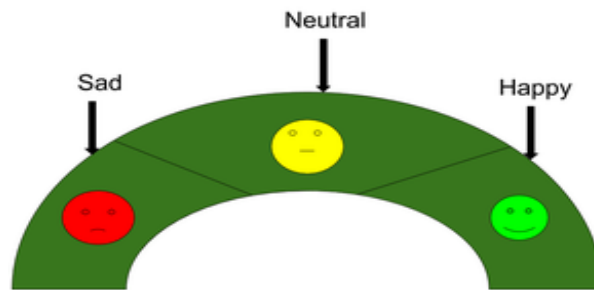


Fig. Sentiment

Steps Involved:

1. Text Preprocessing: Cleaning data (removing punctuation, stop words, etc.)
2. Tokenization: Breaking down sentences into individual words
3. Feature Extraction: Using methods like Bag of Words or TF-IDF
4. Model Training: Using classifiers like Logistic Regression, Naive Bayes, or LSTM
5. Prediction: Classifying new data into positive, negative, or neutral sentiment

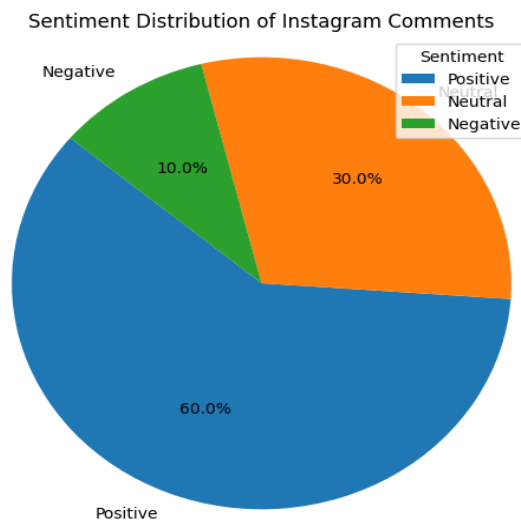
Importance of Sentiment Analysis:

Sentiment analysis is important because it helps us understand the emotions and opinions behind written text. In a world full of online reviews, tweets, and customer feedback, it's impossible to manually read everything — this is where sentiment analysis comes in.

- Understand Customer Opinions: Companies can know what people like or dislike about their products or services.
- Improve Products & Services: Feedback analysis helps improve customer experience.
- Monitor Brand Reputation: Helps track how people feel about a brand in real time.
- Market Research: Businesses use it to analyze trends, preferences, and public mood.
- Decision Making: Provides insights that guide business and marketing strategies.
- Social Media Monitoring: Quickly tracks public reaction to events or campaigns.

Applications of Sentiment analysis:

- Social Media
- Customer Service
- Marketing Sector



Conclusion:

Today's session gave us a hands-on idea of how machines can detect emotions in written text. From companies checking feedback to governments analyzing public mood, sentiment analysis has a wide range of applications. It was interesting to see how data and words come together to reveal insights about people's opinions.